

Adaptive Design of Clustered Experiments



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Motivation

Literature Review

Design Problem under Known ρ

Design Problem under Unknown ρ

Adaptive Algorithm

Lower Bound and Policy Comparison

Simulations

Conclusion

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- ▶ Surveys and experiments have become **larger**
 - ▶ Not just by adding more units, but by spanning **larger geographical regions**
 - ▶ Either physically (provinces, schools) or figuratively (markets, industries)
- ▶ **Why?** [Muralidharan and Niehaus, 2017]
 - ▶ **External Validity.** We want our results to be representative of the population of interest
 - ▶ Get better sense of implementation dynamics
 - ▶ Governmental support

- ▶ Ideally, draw units directly from the population of interest, but
 - ▶ **Logistically infeasible** and **very costly**
 - ▶ Imagine a researcher going all around the country interviewing people
- ▶ **Solution?**
 - ▶ Run **multistage/stratified** surveys and experiments [Sedgwick, 2015], [Barcaroli et al., 2022]
 - ▶ **Structure**
 - ▶ Split the population of interest in *independent* clusters
 - ▶ Randomly select a number of clusters K
 - ▶ Randomly select a set of units N from each of the K clusters
 - ▶ Tower the process if needed

- ▶ **Cluster** related **fixed costs**: partnership with local authorities, travel expenses, additional staff, location specific infrastructure...
- ▶ When there is a hard **budget**, a clear **trade-off** emerges
 - ▶ **number of clusters** (less correlation between observations hence more efficient estimators but also more expensive)
 - ▶ and **number of units** per cluster (higher variance but less expensive)
- ▶ This trade-off is mediated by the degree of **within-cluster correlation**
- ▶ **Research Question**
 - ▶ Can we design optimal (adaptive) multistage sampling experiments under unknown within-cluster correlation?

► Development Economics

- [Mbiti et al., 2019] (QJE) randomly sample 10 districts from mainland Tanzania, and then randomly allocate treatment (combination of school grants and teacher incentives) to 35 schools within each district
- [Karing, 2024] (QJE) select 4 out of Sierra Leone's 14 districts (together with the Ministry of Health and according to some logistic and medical considerations), and then *randomly* allocated treatment (a bracelet system associated to vaccination) to 30 clinics per district
- [Heß et al., 2021] (REStud) analyze the Gambian CDD program (funds at the village level for community development projects). They selected 88 out of 114 wards and then randomized treatment across villages within wards

- ▶ **Survey Design** (prevalent approach)
 - ▶ [Sedgwick, 2015] Survey on British public opinion on the use of personal medical data. Randomly select postcodes from the UK (proportionally to size), and then randomly select households from postcodes
 - ▶ [Osei and Zhuang, 2020] Survey entrepreneurial women in rural Ghana. Randomly sample 15 communities within a district, and then survey an average of 22 female entrepreneurs per community

- ▶ **Stratification** seems **prevalent** in RCT and survey designs, but there are “**no clear guidelines**” driving this process
- ▶ Selection based on *feel*, logistics, or, even worse, **common misconceptions**
 - ▶ Sample **as many clusters as possible**, UN Handbook of Household Sample Surveys in Developing and Transition Economies, [Tam, 2005]
 - ▶ **Fix number of clusters** in advance, [Sedgwick, 2015]
 - ▶ **Guess $\hat{\rho}$** , do minmax or estimate it from previous studies, [Barcaroli et al., 2022]

- ▶ **Framework** for optimal K vs N selection in stratified surveys and experiments under budget constraints and linear costs
- ▶ Characterize the **oracle policy** (and oracle variance) under known correlation
- ▶ Propose an **optimal algorithm** with **unknown cluster correlation**
 - ▶ Show its excess variance is **negligible** compared to the variance of the oracle
 - ▶ Show its excess variance is **unimprovable** through matching lower bounds
 - ▶ Show its excess variance is **non-trivial** when compared to optimal static policies
- ▶ Algorithm uses a **two-stage approach**
 - ▶ **First stage learns correlation** through a stopping algorithm
 - ▶ **Second stage** uses first-stage estimates as **plug-ins for optimization**

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- ▶ Large literature on cluster vs unit selection in **survey design**
[Biemer and Lyberg, 2003], [Bethel, 1989], [Barcaroli et al., 2022], where
 - ▶ They emphasize the role of variable and **cluster costs**
 - ▶ They emphasize the role of **within-cluster correlation** (ICC)
 - ▶ Despite impressive theoretical and algorithmic display featuring complicated objectives, DGP and cost structures, **no attempt at learning ICC**

- ▶ Little econometrics literature on K vs N , but complementary to
 - ▶ **covariate based stratified treatment allocation** [Cytrynbaum, 2021], [Cytrynbaum, 2024], [Bai, 2022], [Bai et al., 2024]
 - ▶ **adaptive experimental design** in economics [Kasy and Sautmann, 2021], [Cesa-Bianchi et al., 2025], [Adusumilli, 2026], [Imbens et al., 2025], [Li et al., 2026]
 - ▶ **classic** literature on (panel data) econometrics with **nested random effects** [Wooldridge, 2003]
 - ▶ bandit literature on **optimal sampling designs** [Lattimore and Szepesvári, 2020], [Kiefer and Wolfowitz, 1960]
- ▶ Use **time-uniform sequences** results in [Howard et al., 2021]

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- ▶ Focus on **survey design** problem. Experimental design on its simplest difference-in-means version is completely equivalent
- ▶ (Nested) **random effects** model

$$y_{ki} = \mu + \varepsilon_k + \varepsilon_{ki}, \quad \mathbb{E}[\varepsilon_k] = \mathbb{E}[\varepsilon_{ki}] = 0, \quad \mathbb{E}[\varepsilon_k^2] = \rho, \quad \mathbb{E}[\varepsilon_{ki}^2] = 1 - \rho$$

- ▶ $\mathbb{E}[y_{ki}] = \mu$
- ▶ equal unit level variance, i.e. $\text{Var}(y_{ki}) = 1$ for all k, i , (*can be relaxed*)
- ▶ **cross-cluster independence**, i.e. $\text{CoVar}(y_{ki}, y_{lj}) = 0$ for all i, j and $k \neq l$,
- ▶ **bounded within-cluster equicorrelation**, i.e. $\text{CoVar}(y_{ki}, y_{kj}) = \rho$ for all i, j , and all k (*hoping to relax*) with $\rho \in [\rho_m, \rho_M] \subset (0, 1)$

► **Estimator**

$$\hat{\mu}(K, N) = \frac{1}{KN} \sum_{k=1}^K \sum_{i=1}^N y_{ki}$$

► **Minimize $\hat{\mu}$ MSE subject to a budget constraint and linear costs**

- $\hat{\mu}$ is unbiased under stratified randomization, so simply minimize variance

$$\begin{aligned} \min_{K \geq 1, N \geq 1} W(K, N; \rho) \\ \text{st } FK + VKN \leq B \end{aligned}$$

► where

- $W(K, N; \rho) = \text{Var}(\hat{\mu}(K, N))$
- F is the **fixed cost** of sampling an additional cluster
- V is the **variable cost** of sampling an additional unit
- $B > F + V$ is the hard **budget**

- Some simplifications

$$\text{Solve } \text{Var}(\hat{\mu}(K, N)) = \frac{1 - \rho}{KN} + \frac{\rho}{K} \quad \text{Solve } K \text{ from BC, } K^*(N) = \frac{B}{F + VN}$$

- The problem reduces to

$$\min_{1 \leq N \leq (B-F)/V} \frac{F + VN}{B} \left(\rho + \frac{1 - \rho}{N} \right)$$

- Define the **optimal mapping** (via FOC)

$$K^*(r) = \frac{B}{F + VN^*(r)}, \quad N^*(r) = \min \left\{ \max \left\{ 1, \sqrt{\frac{F}{V} \frac{1 - r}{r}} \right\}, \frac{B - F}{V} \right\}$$

- ▶ Variance of the optimal mapping $\mathbf{W}^r(\mathbf{r}; \boldsymbol{\rho}) = W(N^*(r), K^*(r); \rho)$
- ▶ Variance of the oracle policy is $\mathcal{O}(1/B)$

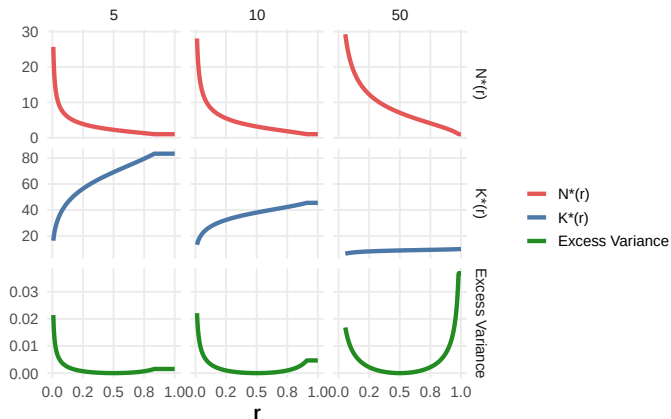
$$\mathbf{W}^r(\boldsymbol{\rho}; \boldsymbol{\rho}) = W(N^*(\rho), K^*(\rho); \rho) = \frac{\left(\sqrt{V(1-\rho)} + \sqrt{F\rho}\right)^2}{B}$$

- ▶ Define the **local excess variance** of the optimal map as

$$W^r(r, \rho) - W^r(\rho, \rho) \approx \frac{1}{2} W_{rr}^r(\rho; \rho) (r - \rho)^2 \sim \mathcal{O}(1/B)$$

- ▶ More generally, for some policy π define its **excess variance**

$$\Delta_{\pi} = \mathbb{E}_{\pi}[W(N_{\pi}, K_{\pi}; \rho)] - W^r(\rho; \rho)$$



$N^*(r)$ (top), $K^*(r)$ (middle) and excess variance $\Delta_\pi(r)$ (bottom) for different F/V ratios.
 $B = 1000, V = 2, \rho = 0.5$

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- Note

$$\text{Var}(\hat{\mu}_k) = \frac{1 + (N-1)\rho}{N} \implies \rho = \frac{N \text{Var}(\hat{\mu}_k) - 1}{N-1}$$

- what suggests the MLE **estimator**

$$\hat{s}^2(K, N) = \frac{1}{K} \sum_{k=1}^K (\hat{\mu}_k(K, N) - \hat{\mu}(K, N))^2 \quad \hat{\rho}(K, N) = \frac{N \hat{s}^2(K, N) - 1}{N-1}$$

- Learning ρ *only* depends on **number of clusters**, not number of units
 - Why? $\text{Var}(\hat{\rho}(K, N))$ is **nearly flat** in N
 - In the paper, I formally characterize the *AVAR* $\hat{\rho}(K, N)$ under K and N asymptotics

- ▶ Say we commit a budget B_0 for a pilot to learn $\hat{\rho}$
 - ▶ “Clusters are all that matters” so set N_0 small (≥ 2) and explore as many clusters as budget allows for $K_0 = B_0/(F + VN_0)$
 - ▶ Estimate $\hat{\rho}(K_0, N_0)$
- ▶ Use remaining budget $B - B_0$ to get as close as possible to $K^*(\hat{\rho}), N^*(\hat{\rho})$, i.e.

$$K_1 = \max\{0, K^*(\hat{\rho}) - K_0\} \quad N_{1k:k \leq K_0} = \max\{0, N^*(\hat{\rho}) - N_0\}$$
$$N_{1k:K_0 \leq k \leq K_1} = N^*(\hat{\rho})$$

- ▶ We won't worry about $N_0 > N^*(\hat{\rho})$, but what if $K_0 > K^*(\hat{\rho})$?
 - ▶ We have **oversampled clusters** and incurred in **irreversible fixed costs**. No way of getting close to $K^*(\rho)$
 - ▶ Mathematically, the error is **not local** to ρ

- ▶ High B_0 gets us a better $\hat{\rho}(N_0, K_0)$, so $(K^*(\hat{\rho}), N^*(\hat{\rho})) \approx (K^*(\rho), N^*(\rho))$
- ▶ But too high B_0 renders the optimal map $(K^*(\hat{\rho}), N^*(\hat{\rho}))$ budget infeasible
- ▶ In summary, we want to set K_0 **as large as possible without compromising the second stage optimization**
- ▶ **Solution?** Run the initial **pilot through a stopping algorithm** that delays stopping as much as possible, but stops before oversampling $k > K^*(\rho)$ (with high probability)

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- We want a **stopping alg** $\mathcal{A} : H_{k-1} \mapsto [0, 1]$ which induces a dist over stopping times τ

$$\mathbb{P}_{\mathcal{A}}(\tau \leq K^*(\rho) \text{ AND } |\hat{\rho}(\tau, N_0) - \rho| \leq D_{\tau}(\alpha)) \geq 1 - \alpha \quad (1)$$

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- Consider an algorithm which, in every period, constructs a CI $C_k(\alpha) = [\hat{\rho}(k, N_0) - D_k(\alpha), \hat{\rho}(k, N_0) + D_k(\alpha)]$ which **contains ρ uniformly with high probability**, i.e.

$$\mathbb{P}(|\hat{\rho}(k, N_0) - \rho| \leq D_k(\alpha) \quad \forall k) \geq 1 - \alpha$$

- If $k + 1 \leq K^*(\hat{\rho} - D_k(\alpha))$ sample an additional cluster, else stop (provided $K^*(r)$ is increasing in r)
- This alg satisfies Eq 1 by design (and stops in finite time wp 1 given $B < \infty$)

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- ▶ If $k + 1 \leq K^*(\hat{\rho} - D_k(\alpha))$ sample an additional cluster, else stop (provided $K^*(r)$ is increasing in r)
 - ▶ This alg satisfies Eq 1 by design (and stops in finite time wp 1 given $B < \infty$)
- ▶ Sadly, **frequentist CI do not satisfy** the time uniformity property

- Instead, consider a cluster summary statistic $\tilde{\mu}_k$ and a 3-diff LL $\mathcal{L}(r \mid \tilde{\mu}_k)$ single-peaked at $r = \hat{\rho}$,

- **Example.** Under normal random effects ε_k and ε_{ki} ,

$$\hat{\mu}_k \sim N(\mu, \sigma^2(\rho)), \quad \sigma^2(\rho) = N_0 + \frac{1 - \rho}{\rho} \quad \mathcal{L}(r \mid \hat{\mu}_k) = \frac{1}{\sqrt{2\pi\sigma^2(r)}} \exp\left(-\frac{(\hat{\mu}_k - \mu)^2}{2\sigma^2(r)}\right)$$

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- For smooth prior $\Pi(r)$ with $\Pi(\rho) > 0$, define the **LR/mixing martingale**

$$M_k(r) := \int \frac{\mathcal{L}(\eta \mid \tilde{\mu}_k)}{\mathcal{L}(r \mid \tilde{\mu}_k)} \Pi(d\eta)$$

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- ▶ Consider the **sequence of confidence intervals**

$$C_k(\alpha) \{r : M_k(r) < 1/\alpha\} \quad \text{where} \quad D_k(\alpha) = \max\{\hat{\rho} - \inf C_k(\alpha), \sup C_k(\alpha) - \hat{\rho}\}$$

► Martingale Interpretation

- Classical LR intuition. If the data looks more like the mixing (numerator) than it does for a candidate r (denominator), then $M_k(r)$ grows (evidence against r) and it is excluded from $\{r : M_k(r) < 1/\alpha\}$

► Theorem 5.1. Anytime Confidence Sequences

$$\mathbb{P}(\rho \in C_k(\alpha) \ \forall k) \geq 1 - \alpha$$

proof

Algorithm ACS

Input $B, \alpha, N_0, \mathcal{L}(\cdot \mid \cdot), \Pi(\eta), \rho_m$

Initialize $k = 0, C_0(\alpha) = [0, 1]$

while $k + 1 \leq \max\{K^*(\rho_m), K^*(\inf C_k(\alpha))\}$ and $k + 1 \leq \frac{B}{F+VN_0}$

explore an additional cluster $k = k + 1$ and **observe** $(y_{ki})_{N_0}$

compute the sufficient statistic $\tilde{\mu}_k$

update $\mathcal{L}(\cdot \mid \tilde{\mu}_k), M_k(r), C_k(\alpha)$

set $K_0 = k$ and **compute** $\hat{\rho}(K_0, N_0)$

sample $K_1 = \max\{0, K^*(\hat{\rho}) - K_0\}$

$$N_{1k} = N_{1k:k \leq K_0} = \max\{0, N^*(\hat{\rho}) - N_0\}, \quad N_{1k:K_0 \leq k \leq K_1} = N^*(\hat{\rho})$$

- Define the **good event** $\mathcal{E} = \{\rho \in C_k(\alpha) \ \forall k\}$, $\mathbb{P}(\mathcal{E}) \geq 1 - \alpha$ (Thm 5.1)

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- ▶ **Lemma 5.2. Algorithm Performance**

$$\Delta_{\text{ACS}} \mid \mathcal{E} \leq \frac{1}{2} W_{rr}^r(\rho, \rho) \mathbb{E}_{\text{ACS}}[D_\tau(\alpha)^2 \mid \mathcal{E}] + R_\tau$$

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- **Lemma 5.3. Upper Bound on $D_k(\alpha)$**

$$D_k(\alpha) \leq \sqrt{\frac{2 \ln(1/\alpha) + C \ln k}{k I(\hat{\rho}_k)}} (1 + \delta_k) \text{ for some seq } \delta_k \rightarrow 0$$

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- ▶ **Lemma 5.4. Linear Stopping Time**

for all sufficiently large B , $\mathbb{P}_{\text{ACS}}(\tau \in \Theta(B) \mid \mathcal{E}) = 1$

Theorem 5.5. Upper Bound on ACS

Under the assumptions in Lemmas 5.2, 5.3 and 5.4 and $0 < I_m < I(r) < \infty$, wp $1 - \alpha$, there exists a constant $C < \infty$ such that

$$\Delta_{\text{ACS}} \lesssim C \cdot \frac{\ln(1/\alpha) + \ln B}{B^2}$$

- The excess variance of ACS is **asymptotically negligible** compared to the oracle variance $O(1/B) \gg O(\ln B/B^2)$

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- ▶ Show that ACS excess variance is **essentially unimprovable** by deriving **matching lower bounds** on the problem up to logarithmic terms
- ▶ **Proof Structure**
 - ▶ Consider two DGPs indexed by $\bar{\rho} = \rho + \delta$ and $\underline{\rho} = \rho - \delta$ with $\delta \sim O(B^{-1/2})$
 - ▶ δ is **small enough** such that it is difficult to distinguish across instances
 - ▶ δ is **large enough** such that misidentification is costly
 - ▶ Finally, argue that any policy which does well under $\bar{\rho}$ must not do very well under $\underline{\rho}$ and viceversa

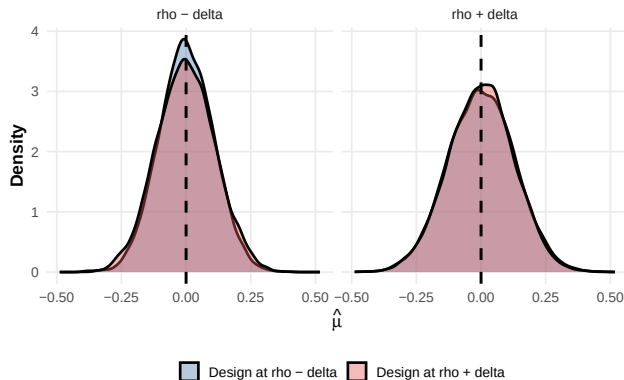


Figure: Distribution of $\hat{\mu}(K, N)$ under $\underline{\rho} = \rho - \delta$ (left) and $\bar{\rho} = \rho + \delta$ (right). $\rho = 0.3$, $\delta = 0.1$, $B = 1000$, $F = 20$, $V = 2$. 10,000 replications.

Theorem 6.1. Lower Bound on Excess Variance

For any policy π , such that $P_\pi \left(\sum_k^K (F + V N_k) \leq B \right) = 1$ there exists a DGP indexed by $\rho \in (\rho_m, \rho_M)$ such that

$$\Delta_\pi \gtrsim \frac{1}{B^2} \frac{V}{44\rho} \sqrt{\frac{FV(1-\rho)}{\rho}} \sim O(1/B^2)$$

- ACS is **nearly-optimal** up to logarithmic and constant terms

- ▶ The excess variance of ACS $O(\ln B/B^2)$ is **not trivial**
- ▶ **Lemma 6.2.**
 - ▶ Any static policy which takes a **mispecified** $\tilde{\rho} = \rho + b$,
 - ▶ and the **minmax optimal** policy over $\rho \in [\rho_m, \rho_M]$
 - ▶ accrue an excess variance of **$O(1/B)$** $\gg O(\ln B/B^2)$

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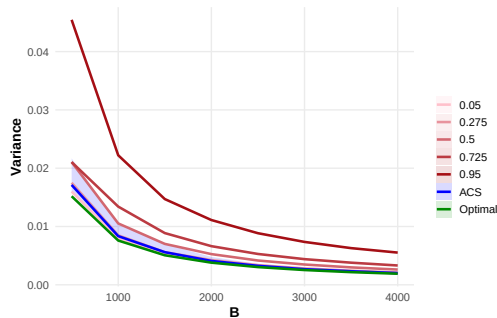
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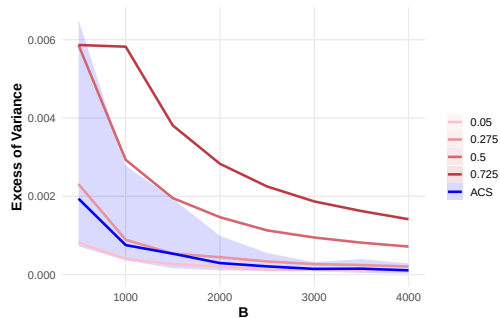
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Policy	K	N	Share $K_0 > K^*(\rho)$	Δ_π	$Q_{0.05}(\Delta_\pi)$	$Q_{0.95}(\Delta_\pi)$
Oracle	77	9	0	0	-	-
ACS	77	9.5	.03	.00017	.000060	.00042
Fixed .05	60	15	0	.00014	-	-
Fixed .275	100	5	1	.00027	-	-
Fixed .5	115	3	1	.00095	-	-
Fixed .725	125	2	1	.0019	-	-
Fixed .95	136	1	1	.0048	-	-

Table: Performance Comparison across Policies $\rho = 0.1$. $\alpha = 0.05$, $B = 3000$, $F = 20$, $V = 2$. 100 replications.



(a) Variance $W(N_{\text{ACS}}, K_{\text{ACS}}; \rho)$



(b) excess variance $W(N_{\text{ACS}}, K_{\text{ACS}}; \rho) - W^r(\rho, \rho)$

Figure: Variance (left) and Excess Variance (right) as a Function of B across Policies.
 $\rho = 0.1$, $\alpha = 0.05$, $F = 20$, $V = 2$. 100 replications.



Figure: K_{0ACS} and K_{ACS} (histogram left), and $K^*(\rho)$ (line plot right). $\rho = 0.5$, $\alpha = 0.05$, $B = 2000$, $F = 20$, $V = 2$. Dashed line equals $K^*(0.5)$.

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- ▶ Proposed a **framework** for optimal K vs N selection in stratified surveys and experiments under budget constraints and linear costs
- ▶ Characterized the **oracle policy** (and oracle variance) under known correlation
- ▶ Proposed an **optimal algorithm** with **unknown cluster correlation**
 - ▶ Showed its excess variance is **negligible** compared to the variance of the oracle
 - ▶ Showed its excess variance is **unimprovable** through matching lower bounds
 - ▶ Showed its excess variance is **non-trivial** when compared to optimal static policies

Thank you!

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- By (simplified) Ville's Inequality, for any non-negative martingale M_k and any real $\alpha > 0$

$$\mathbb{P} \left(\exists k : M_k \geq \frac{1}{\alpha} \right) \leq \alpha \mathbb{E}[M_0]$$

- Thus, it remains to show that M_k is non-negative martingale with $\mathbb{E}[M_0] = 1$

$$\begin{aligned} \mathbb{E}_r \left[\prod_k \frac{f_\eta(\tilde{\mu}_k)}{f_r(\tilde{\mu}_k)} \right] &= \prod_k \mathbb{E}_r \left[\frac{f_\eta(\tilde{\mu}_k)}{f_r(\tilde{\mu}_k)} \right] && \text{cross-cluster independence} \\ &= \prod_k \int \frac{f_\eta(\tilde{\mu}_k)}{f_r(\tilde{\mu}_k)} f_r(\tilde{\mu}_k) d\tilde{\mu}_k && \text{definition of } \mathbb{E}_r \\ &= \prod_k \int f_\eta(\tilde{\mu}_k) d\tilde{\mu}_k = 1 && \text{valid density} \end{aligned}$$

► Therefore,

$$\mathbb{E}_r \left[\int \prod_k \frac{f_\eta(\tilde{\mu}_k)}{f_r(\tilde{\mu}_k)} \Pi(d\eta) \right] \stackrel{\text{Fub}}{=} \int \mathbb{E}_r \left[\prod_k \frac{f_\eta(\tilde{\mu}_k)}{f_r(\tilde{\mu}_k)} \right] \Pi(d\eta) = \int 1 \Pi(d\eta) = 1$$

return